

CLASS 9 · CBSE · GANITA MANJARI PART 1 · SHORT NOTES

Chapter 8: Predicting What Comes Next? Exploring Sequences and Progressions

Topic-wise short notes with solved examples

A sequence is an ordered list of numbers. This chapter shows two ways of writing its rule, and studies two special sequences: the arithmetic progression, which grows by adding, and the geometric progression, which grows by multiplying.

1 Sequences and their terms

- A **sequence** is an ordered list of numbers. Each number is a **term**.
- A sequence is **finite** if it stops (6, 12, 24, 48, 96) and **infinite** if it goes on for ever. Three dots show that it continues.
- The terms are written $t_1, t_2, t_3, \dots$. The subscript is the position: $t_4 = 7$ means the fourth term is 7. The position n is always a natural number, but the term itself can be any number.

Sequence	Terms	Pattern
Natural numbers	1, 2, 3, 4, 5, ...	each term is 1 more than the last
Odd numbers	1, 3, 5, 7, 9, ...	the difference is always 2
Triangular numbers	1, 3, 6, 10, 15, ...	1, 1 + 2, 1 + 2 + 3, ...
Square numbers	1, 4, 9, 16, 25, ...	1, 1 + 3, 1 + 3 + 5, ...

- The n th square number is the sum of the first n odd numbers. The n th triangular number is the sum of the first n natural numbers.
- Some sequences, such as the prime numbers 2, 3, 5, 7, 11, ..., have no simple rule.

2 Explicit rule

- An **explicit rule** gives the n th term directly from its position n . Any term can be found without knowing the earlier ones.
- To check whether a number is a term, set the rule equal to the number and solve for n . It is a term only if n is a natural number.

Example 2.1 For the sequence $t_n = 4n + 1$, find the first three terms and the 25th term. Are 213 and 150 terms of the sequence?

SOLUTION

$$t_1 = 5, t_2 = 9, t_3 = 13 \text{ and } t_{25} = 101$$

$$4n + 1 = 213 \text{ gives } n = 53. 4n + 1 = 150 \text{ gives } n = 37.25.$$

5, 9, 13; $t_{25} = 101$; 213 is the 53rd term; 150 is not a term

3 Recursive rule

- A **recursive rule** gives the first term (or first few terms) and a way to get each term from the terms before it.
- To use it, the earlier terms must be worked out first.

Example 3.1 Find the first four terms of the sequence $u_1 = 2$, $u_n = 3u_{n-1} - 1$ for $n \geq 2$.

SOLUTION

$$u_2 = 3(2) - 1 = 5, u_3 = 3(5) - 1 = 14, u_4 = 3(14) - 1 = 41$$

2, 5, 14, 41

- The **Virahāṅka-Fibonacci sequence** uses the two previous terms: $V_1 = 1$, $V_2 = 2$ and $V_n = V_{n-1} + V_{n-2}$ for $n \geq 3$. It gives 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, ...

REMEMBER

Virahāṅka wrote down this sequence in the 7th century CE while studying the metres of Prakrit poetry. Gopāla and Hemachandra studied it further, well before Fibonacci (c. 1200 CE).

4 Arithmetic progressions

- An **arithmetic progression (AP)** is a sequence in which each term is obtained by **adding** a fixed number to the term before it. The fixed number is the **common difference** d .
- General form: a , $a + d$, $a + 2d$, $a + 3d$, ... where a is the first term.

$$t_n = a + (n - 1)d$$

- Recursive rule: $t_1 = a$ and $t_n = t_{n-1} + d$ for $n \geq 2$.
- d can be negative: 11, 7, 3, -1, ... is an AP with $d = -4$.
- Plotting the points (n, t_n) of an AP gives points on a **straight line**. An AP is a linear pattern.

Example 4.1 Find the n th term and the 20th term of the AP 7, 11, 15, 19, ...

SOLUTION

$$a = 7 \text{ and } d = 4, \text{ so } t_n = 7 + (n - 1) \times 4 = 4n + 3.$$

$$t_n = 4n + 3; t_{20} = 83$$

Example 4.2 Which term of the AP 20, 17, 14, ... is -40? Is 0 a term of this AP?

SOLUTION

$$t_n = 20 + (n - 1)(-3) = 23 - 3n$$

$$23 - 3n = -40 \text{ gives } n = 21. \quad 23 - 3n = 0 \text{ gives } n = \frac{23}{3}, \text{ which is not a natural number.}$$

-40 is the 21st term; 0 is not a term

Example 4.3 The 4th term of an AP is 17 and the 9th term is 37. Find the AP and its 15th term.

SOLUTION

$a + 3d = 17$ and $a + 8d = 37$. Subtracting: $5d = 20$, so $d = 4$ and $a = 5$.

$$t_{15} = 5 + 14 \times 4$$

The AP is 5, 9, 13, ... and $t_{15} = 61$

Example 4.4 How many three-digit numbers are divisible by 6?

SOLUTION

They form the AP 102, 108, ..., 996 with $d = 6$.

$$996 = 102 + (n - 1) \times 6, \text{ so } n - 1 = 149.$$

150 numbers

5 Sum of the first n natural numbers

- Write the sum forwards and backwards and add:

$$S = 1 + 2 + \dots + n \quad S = n + (n - 1) + \dots + 1$$

- Each of the n pairs adds up to $n + 1$, so $2S = n(n + 1)$.

$$S_n = 1 + 2 + 3 + \dots + n = \frac{n(n + 1)}{2}$$

- Āryabhaṭa stated this result in the Āryabhaṭīya (499 CE): the sum is the average of the first and last terms multiplied by the number of terms.
- The n th **triangular number** is $\frac{n(n + 1)}{2}$.
- A sum that does not start from 1 is a difference of two such sums.

Example 5.1 Find $1 + 2 + 3 + \dots + 40$ and the 12th triangular number.

SOLUTION

$$S_{40} = \frac{40 \times 41}{2} \text{ and } \frac{12 \times 13}{2}$$

820 and 78

Example 5.2 Find $31 + 32 + 33 + \dots + 70$.

SOLUTION

$$\text{The sum is } S_{70} - S_{30} = \frac{70 \times 71}{2} - \frac{30 \times 31}{2} = 2485 - 465$$

2020

6 Geometric progressions

- A **geometric progression (GP)** is a sequence in which each term is obtained by **multiplying** the term before it by a fixed number. The fixed number is the **common ratio** r .
- General form: $a, ar, ar^2, ar^3, \dots$

$$t_n = ar^{n-1}$$

- Recursive rule: $t_1 = a$ and $t_n = r \times t_{n-1}$ for $n \geq 2$.
- To test for a GP, check that the ratio of each term to the one before it is the same.
- r can be a fraction or negative: $1, -1, 1, -1, \dots$ is a GP with $r = -1$.
- Plotting the points (n, t_n) of a GP does **not** give a straight line. The points rise (or fall) faster and faster.

Example 6.1 Is 8, 12, 18, 27, ... a GP? If so, find its n th term.

SOLUTION

$$\frac{12}{8} = \frac{3}{2}, \frac{18}{12} = \frac{3}{2}, \frac{27}{18} = \frac{3}{2}. \text{ The ratio is constant.}$$

Yes, with $a = 8$ and $r = \frac{3}{2}$: $t_n = 8 \left(\frac{3}{2}\right)^{n-1}$

Example 6.2 Find the 7th term of the GP 2, 6, 18, ... Which term of the GP 5, 10, 20, ... is 1280?

SOLUTION

$$t_7 = 2 \times 3^6 = 2 \times 729$$

$$5 \times 2^{n-1} = 1280, \text{ so } 2^{n-1} = 256 = 2^8 \text{ and } n = 9.$$

1458; the 9th term

Example 6.3 A culture starts with 50 bacteria and the number doubles every hour. How many are there after 6 hours and after n hours?

SOLUTION

After each hour the number is multiplied by 2, so after n hours it is 50×2^n .

3200 after 6 hours; 50×2^n after n hours

Example 6.4 A ball is dropped from a height of 50 m and each bounce reaches 80% of the previous height. How high does it rise after the 4th bounce?

SOLUTION

The heights form a GP with first term $50 \times 0.8 = 40$ and $r = 0.8$: 40, 32, 25.6, 20.48.

20.48 m

7 Fractals and geometric progressions

- A **fractal** is a pattern that repeats itself at smaller and smaller scales.
- Sierpiński triangle**: start with a triangle (stage 0). At each stage, join the mid-points of the sides of every shaded triangle and remove the middle one.

Stage n	0	1	2	3	n
Number of shaded triangles	1	3	9	27	3^n
Shaded area (stage 0 has area 1)	1	$\frac{3}{4}$	$\frac{9}{16}$	$\frac{27}{64}$	$\left(\frac{3}{4}\right)^n$

- The number of triangles is a GP with ratio 3 and grows quickly. The shaded area is a GP with ratio $\frac{3}{4}$ and gets closer and closer to 0.

8 AP and GP compared

	AP	GP
How the next term is found	add d	multiply by r
What stays constant	difference $t_n - t_{n-1}$	ratio $\frac{t_n}{t_{n-1}}$
n th term	$a + (n - 1)d$	ar^{n-1}
Graph of (n, t_n)	points on a straight line	points not on a straight line

9 Common mistakes

- Writing $t_n = a + nd$. The first term has no d added, so it is $a + (n - 1)d$.
- Writing the n th term of a GP as ar^n . The power is $n - 1$.
- Accepting a value of n that is not a natural number. Then the number is not a term.
- Taking the common difference as (earlier term – later term). It is $t_2 - t_1$.
- Calling a sequence a GP after checking only one ratio. Check each pair of consecutive terms.